A First Course in Abstract Algebra

Appendix: Matrix Algebra

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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A Appendix: Matrix Algebra

An $m \times n$ matrix has m rows and n columns. Addition and subtraction require the same dimensions and operate entry by entry. The product of an $m \times n$ matrix and an $n \times r$ matrix is an $m \times r$ matrix: its (i, j) entry is the dot product of row i with column j. Throughout the complex calculations, use $i^2 = -1$.

Exercise A.1: Adding two real matrices

Plan. Add entries occupying the same position; both matrices are 2×2.

$$\begin{bmatrix} -2 & 4 \\ 1 & 5 \end{bmatrix} + \begin{bmatrix} 4 & -3 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} -2+4 & 4-3 \\ 1+1 & 5+2 \end{bmatrix} = \boxed{\begin{bmatrix} 2 & 1 \\ 2 & 7 \end{bmatrix}}.$$

No row is multiplied by a column when adding matrices.

Exercise A.2: Adding two complex matrices

Both matrices are 2×3, so addition is defined. Add real parts to real parts and imaginary parts to imaginary parts:

$$(1+i)+3 = 4+i, \qquad -2+(i-1) = -3+i, \qquad (3-i)+(-2+i) = 1,$$
$$4+(3-i) = 7-i, \qquad i+(1+i) = 1+2i, \qquad (2-i)+0 = 2-i.$$

Conclusion. The sum is $\begin{bmatrix} 4+i & -3+i & 1 \\ 7-i & 1+2i & 2-i \end{bmatrix}$.

Exercise A.3: Subtracting complex matrices

Subtraction is defined because both matrices are 3×2. Distribute each minus sign to the entire corresponding entry:

$$\begin{bmatrix} i & -1 \\ 4 & 1 \\ 3 & -2i \end{bmatrix} - \begin{bmatrix} 3-i & 4i \\ 2 & 1+i \\ 3 & -i \end{bmatrix} = \begin{bmatrix} i-(3-i) & -1-4i \\ 4-2 & 1-(1+i) \\ 3-3 & -2i-(-i) \end{bmatrix} = \boxed{\begin{bmatrix} -3+2i & -1-4i \\ 2 & -i \\ 0 & -i \end{bmatrix}}.$$

In particular, subtracting $-i$ adds i.

Exercise A.4: A product of two square matrices

Plan. The inner dimensions match. Compute four row–column dot products.

$$\begin{bmatrix} 1 & -1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 4 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 1(2)+(-1)(-1) & 1(4)+(-1)(3) \\ 3(2)+1(-1) & 3(4)+1(3) \end{bmatrix} = \boxed{\begin{bmatrix} 3 & 1 \\ 5 & 15 \end{bmatrix}}.$$

The factors and the result are all 2×2.

Exercise A.5: A rectangular matrix product

A 2×2 matrix times a 2×3 matrix gives a 2×3 result:

$$\begin{bmatrix} 3 & 1 \\ -4 & 2 \end{bmatrix} \begin{bmatrix} 1 & 5 & -3 \\ 2 & 1 & 6 \end{bmatrix} = \begin{bmatrix} 3(1)+1(2) & 3(5)+1(1) & 3(-3)+1(6) \\ -4(1)+2(2) & -4(5)+2(1) & -4(-3)+2(6) \end{bmatrix} = \boxed{\begin{bmatrix} 5 & 16 & -3 \\ 0 & -18 & 24 \end{bmatrix}}.$$

Exercise A.6: Checking dimensions before multiplying

The first matrix is 2×2 and the second is 3×2. The number of columns of the first, namely 2, differs from the number of rows of the second, namely 3. For example, the proposed first entry would require a dot product between a two-entry row and a three-entry column.

Conclusion. The indicated matrix product is undefined.

Exercise A.7: A complex matrix product

Multiply rows by columns, remembering $i(3i) = 3i^2 = -3$:

$$\begin{bmatrix} i & 1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 3i & 1 \\ 4 & -2i \end{bmatrix} = \begin{bmatrix} i(3i)+1(4) & i(1)+1(-2i) \\ (-2)(3i)+1(4) & (-2)(1)+1(-2i) \end{bmatrix}$$
$$= \boxed{\begin{bmatrix} 1 & -i \\ 4-6i & -2-2i \end{bmatrix}}.$$

Exercise A.8: The fourth power of a real matrix

Let $M = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$. Matrix powers mean repeated matrix multiplication, not taking powers of the individual entries. First compute

$$M^2 = \begin{bmatrix} 1-1 & -1+0 \\ 1+0 & -1+0 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix}.$$

Now square that result:

$$M^4 = (M^2)^2 = \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix} = \boxed{\begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix}}.$$

As a check, multiplying M^2 by M gives $M^3 = -I_2$, so $M^4 = -M$.

Exercise A.9: The fourth power of a complex matrix

Put $N = \begin{bmatrix} 1 & -i \\ i & 1 \end{bmatrix}$. Then

$$N^2 = \begin{bmatrix} 1+(-i)i & -i-i \\ i+i & i(-i)+1 \end{bmatrix} = \begin{bmatrix} 2 & -2i \\ 2i & 2 \end{bmatrix} = 2N.$$

Consequently $N^4 = (N^2)^2 = (2N)^2 = 4N^2 = 8N$, and therefore

$$\boxed{N^4 = \begin{bmatrix} 8 & -8i \\ 8i & 8 \end{bmatrix}}.$$

Only the scalar factors commute here; no interchange of matrix factors is needed.

Exercise A.10: An explicit failure of commutativity

Take integer matrices

$$P = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad Q = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Their products are

$$PQ = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \qquad QP = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

The upper-right entries differ, so $PQ \neq QP$. Both factors belong to $M_2(\mathbb{Z})$ and both products are defined, as required.

Exercise A.11: Inverting a signed permutation matrix

Let $J = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$. Multiplying gives $J^2 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = -I_2$. Thus $J(-J) = (-J)J = I_2$. An inverse must produce the identity on both sides, which proves

$$\boxed{J^{-1} = -J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}}.$$

Exercise A.12: Inverting a diagonal matrix

For $D = \operatorname{diag}(2, 4, -1)$, choose $E = \operatorname{diag}(1/2, 1/4, -1)$. Products of diagonal matrices multiply their corresponding diagonal entries and have zero off-diagonal entries. Hence

$$DE = ED = \operatorname{diag}(2 \cdot \tfrac{1}{2}, 4 \cdot \tfrac{1}{4}, (-1)(-1)) = I_3.$$

This verifies directly that

$$\boxed{D^{-1} = \begin{bmatrix} 1/2 & 0 & 0 \\ 0 & 1/4 & 0 \\ 0 & 0 & -1 \end{bmatrix}}.$$

The inverse is taken over $\mathbb{R}$ or $\mathbb{C}$; it need not have integer entries.

Exercise A.13: The determinant of a triangular matrix

The given matrix is lower triangular: all entries above its main diagonal are zero. Its determinant is therefore the product of its diagonal entries:

$$\det \begin{bmatrix} 3 & 0 & 0 \\ 10 & -2 & 0 \\ 4 & 17 & 8 \end{bmatrix} = 3(-2)(8) = \boxed{-48}.$$

Alternatively, expanding along the first row gives $3 \det \begin{bmatrix} -2 & 0 \\ 17 & 8 \end{bmatrix} = 3(-16)$, the same answer.

Exercise A.14: Products of invertible matrices

What we must show. Given invertible $A, B \in M_n(\mathbb{C})$, exhibit two-sided inverses for AB and BA.

Plan. Reverse the order of the inverses. Matrix multiplication is associative, which allows regrouping without changing the order of any factors.

Check the inverse of AB.

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AI_nA^{-1} = I_n,$$
$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}I_nB = I_n.$$

Thus $(AB)^{-1} = B^{-1}A^{-1}$.

Check the inverse of BA.

Similarly,

$$(BA)(A^{-1}B^{-1}) = B(AA^{-1})B^{-1} = I_n, \qquad (A^{-1}B^{-1})(BA) = A^{-1}(B^{-1}B)A = I_n.$$

Conclusion. Both products are invertible, with $(AB)^{-1} = B^{-1}A^{-1}$ and $(BA)^{-1} = A^{-1}B^{-1}$.